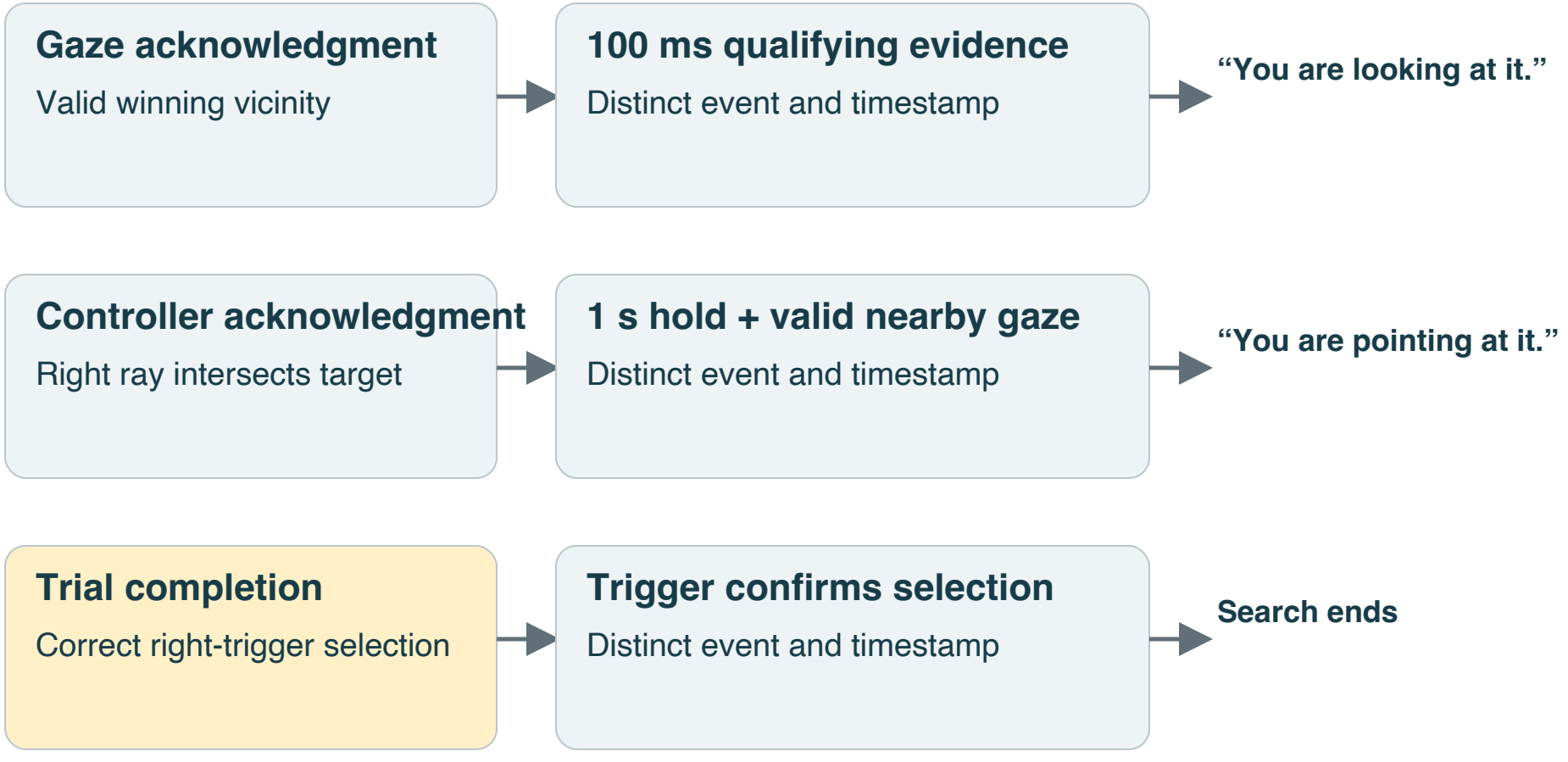
Three different events

Implemented interaction gates; neither ray contact nor vicinity qualification establishes fixation.